

Title

Effectiveness of an integrative exercise and health education program for metabolic syndrome: protocol for a randomized comparative trial in a Latin American community

Abstract

Metabolic syndrome represents a significant public health challenge throughout Latin America, yet evidence-based community interventions remain underutilized in clinical practice. This protocol describes an 18-week structured program designed to improve cardiometabolic health through supervised aerobic activity, evidence-based nutrition and lifestyle education, and regular clinical monitoring. A single-center randomized design will enroll approximately 80 adults with confirmed metabolic syndrome from community health centers. Participants will be randomly allocated to either the integrated exercise-and-education program or standard care. The intervention comprises three supervised aerobic sessions weekly, monthly group health-education workshops, and biweekly clinical assessments including weight and metabolic measures. Primary outcomes examine changes in cardiorespiratory fitness, body composition measures, and metabolic markers including lipids and glucose. Secondary assessments will measure functional capacity, self-reported health status, and adherence rates. Data collection occurs at baseline and 18-week completion. This pragmatic protocol outlines a scalable community model for metabolic syndrome management in resource-limited settings.